A-A-50975
14 June 1985

COMMERCIAL ITEM DESCRIPTION

STIRRER-HOT PLATE, MAGNETIC, LABORATORY

The General Services Administration has authorized the use of this commercial item description in preference to Military Specification MIL-S-36963A.

This commercial item description covers one type of magnetic stirrer-hot plate for use in laboratory procedures.

Salient characteristics:

Shall be a magnetic stirrer-hot plate, suitable for use in laboratory procedures on 115 volts, 50-60 Hz, AC supply lines.

Shall be supplied with a teflon coated magnetic stirring bar with a nominal one or two inch length, or other intermediate length.

Shall be capable of stirring and heating simultaneously or of doing either operation alone. The stirring and heating controls shall have separate pilot lights, and separate on-off switches or integral controls with "off" positions. The pilot light for the hot plate may be used to indicate heat cycling.

Shall be capable of stirring liquids in the viscosity range of 1 to 1500 centipose using the magnetic stirring bar to be supplied. Shall have variable stirring speeds from slow to fast (approximately 950 rpm). The stirring speed for the minimum speed setting shall be approximately 250 rpm or less.

Shall permit selection and control of hot plate temperatures up to approximately 750 degrees (F).

Shall have at maximum control settings a power consumption of 600 to 865 watts.

Shall have a rectangular top surface area of at least 35 square inches with length and width dimensions that are at least 5 inches.

Shall have maximum overall dimensions of 11 long by 8-1/4 deep by 5-1/2 high (inches).

Shall as a minimum consist of the following:

1. A metal base with an epoxy, vinyl or baked enamel finish on the exposed surfaces.
2. A top plate fabricated from glass ceramic material or corrosion-resisting steel coated with porcelain enamel material, having on the exposed surface a glazed finish that is smooth, hard, scratch resistant, chemically inert, and does not become permanently discolored because of having been in a heated condition.
3. A lubrication-free motor assembly with a magnet attached to the drive shaft. Shall include a shaded pole, high torque variable speed motor, for which a 5/6 change in motor speed at 50 Hz operation (i.e., from that at 60 Hz) shall be acceptable.
4. A motor speed control with reference graduation markings, that allows variable stepless speeds.

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5. A heating element.
6. A heater control with reference graduation markings, that includes an adjustable thermostat to permit selection and control of the top plate temperatures.
7. Separate on-off switches or integral controls with "off" positions for the stirring and heater controls.
8. Separate pilot or cycle indicating lights for the stirring and heater controls.
9. A three (3) wire electric power cord with a three (3) blade male connector fitting receptacles described in ANSI C73-11.

Shall conform to the applicable requirements of Underwriters Laboratories' UL 1262 (Standard for Safety - Laboratory Equipment), or UL499 (Standard for Safety-Electric Heating and Appliances), with proof of conformance consisting of one of the following:

1. UL label or UL listing of the product, provided a copy of the UL report is available to the procurement officer, upon the latter's request.
2. Laboratory report from an independent laboratory, provided a copy is available to the procurement officer upon the latter's request. The report shall include actual test methods, data taken, and test results.

Shall be free of rocking and wobbling when placed on a flat surface.

Workmanship. The stirrer-hot plate shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One stirrer-hot plate, as specified, constitutes one unit.

Quality Assurance Provision

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspections. Inspections, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Each stirrer-hot plate shall be tested to demonstrate its capability for proper operation.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U. S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to rapplicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

The following National Stock Number is covered by this document:

6640-00-113-8336

Ordering data (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

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MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

CIVIL AGENCY COORDINATING ACTIVITY:

VA-OSS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA-DM

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Project No. 6640-1203

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